



Renewable malic acid-based plasticizers for both PVC and PLA polymers

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ABSTRACT

Acetylated malic acid alkyl esters (AcMAE-Cn) with different alkyl chain lengths, as alternative plasticizers for both poly(vinyl chloride) (PVC) and poly(lactic acid) (PLA), were synthesized by the esterification of renewable malic acid with alcohols and subsequent acetylation. The molecular structure of the synthesized plasticizers was confirmed using ¹H NMR and FT-IR spectroscopy. PVC or PLA blends containing 50 phr of the plasticizers were evaluated by dynamic mechanical analysis (DMA), tensile testing, and migration testing to evaluate the plasticization efficiency. The plasticized PVC films presented good plasticizing performance, as demonstrated by the results of tensile and migration testing and the *T_g* values (as determined by DMA). Among the plasticizers, AcMAE-C6 and AcMAE-C8 presented enhanced plasticizing performance compared with that of the commercial dioctyl adipate (DOA) and di(2-ethylhexyl) phthalate (DEHP) plasticizers. Additionally, the plasticizing properties of the plasticized PLA films depend significantly on the structure of the plasticizers. It was found that plasticizers (AcMAE-C4 and AcMAE-C6) bearing short alkyl groups (n-butyl and n-hexyl, respectively) may act as excellent plasticizers. In summary, the AcMAE-C6 plasticizer showed potential as an alternative plasticizer for both PVC and PLA polymers.

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Introduction

Poly(vinyl chloride) (PVC) is one of the most widely used polymers in industrial fields such as the manufacturing of packaging, toys, biomedical products, and building materials [1], but its use is generally limited due to the brittle nature of the polymer. Thus, it is often compounded with plasticizers to enhance its flexibility and toughness for various applications. Some of the most common plasticizers that are currently being utilized for PVC are aromatics and aliphatic esters, typically di(2-ethylhexyl) phthalate (DEHP) and dioctyl adipate (DOA). There have been concerns over the use of DEHP, as it negatively impacts

human health and the environment [2]. More specifically, DEHP is a known endocrine disruptor in animals and suspected to also affect humans, in which it could potentially damage not only reproductive function but also immune function, behavior and memory [3]. Many alternative plasticizers have been developed with the discontinuation or restriction of DEHP use in European nations and elsewhere [4–6]. These alternatives should have similar or superior mechanical properties to phthalates in order to function as suitable replacements. Ideally, alternative plasticizers should also be biodegradable and have less propensity than phthalates to leach into the surroundings in order to minimize their negative environmental and health effects. There have already been many studies performed on biobased plasticizers for PVC, such as succinate [7], maleate [8], fumarate [9], vegetable derivatives [10–12], and lactic acid [13], in hopes of dethroning phthalates as a popular plasticizer. However, the environmentally friendly plasticizers developed so far may have problems competing with phthalates in terms of cost as well as plasticization efficiency.

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